Radar-Informed Candidate Detection and Ensemble Scoring for Continuous Ultra-Wideband Fall Detection

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Abstract

Continuous fall detection requires detecting the events and estimating their temporal boundaries in streams dominated by daily activities. This study investigates whether radar-informed proposal generation and compact ensemble scoring can improve event localization without training a dense temporal boundary network. We present an offline pipeline for three ultra-wideband radars that combines exponential background removal, per-stream energy normalization, adaptive proposal refinement, final boundary centering, and a 26-dimensional feature representation. Features describe temporal energy, pre-event and post-event context, changes in channel impulse response profiles, and spatial dispersion over 184 complex range bins. A weighted ensemble of histogram gradient boosting, extremely randomized trees, and a multilayer perceptron scores proposals, followed by suppression independently within each stream and a duration-dependent adjustment of the final interval.The final predictions on the ABC2027 validation split achieve a mean event F1 of 0.8172 over temporal intersection-over-union thresholds of 0.1, 0.3 and 0.5. At threshold 0.3, the precision is 0.9123, the recall is 0.7761 and the false-alarm rate is 2.44 per hour. Median absolute start and end errors are 0.18 and 0.29 seconds, respectively. These results exceed the official energy baseline, although validation-based parameter selection and synthetic stream construction limit generalization claims. The contribution is an explicit proposal-and-scoring formulation for this continuous-stream benchmark; neither real-time alarm performance nor labeled test performance is established.

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1 Introduction

Contactless activity monitoring requires interpreting sensor measurements over time rather than assigning labels only to isolated recordings. For fall detection, a useful system must distinguish falls from daily activities and identify the interval during which each fall occurs. Ultra-wideband (UWB) radar offers measurements without collecting optical images or requiring a worn sensor. Published UWB studies have investigated convolutional and recurrent networks, classical machine learning, transfer learning, and hybrid neural architectures for identifying falls [6, 5, 3, 2]. Their findings motivate further investigation of radar representations, but classification results do not directly establish localization performance on continuous streams.

This is important because an interval may contain a fall but provide a bad estimate of the fall start and end. A broad prediction may be correct about the activity but overlap poorly with the annotated event. In contrast, a prediction that is too short may miss the beginning or settling phase. Repeated responses to ordinary motion also build up as false alarms in a long recording. Evaluation should consider joint detection, temporal overlap, boundary error, and observation duration. These requirements would be obscured by reporting accuracy alone.

The ABC2027 UWB challenge provides a framework to investigate this issue using continuous streams generated from fall and daily activity recordingsciteabc2027. The detector is provided with unsegmented radar data and predicts scored fall intervals. The benchmark ranks predictions by mean event F1 over three temporal overlap thresholds. Provided that streams are processed independently and repeated activity content is not found through matching or alignment, its rules allow the processing of an entire stream. This allows contextual analysis while keeping a clear separation between inference based on signals and exploitation of stream construction.

This study asks whether physics-motivated signal summaries, explicit proposal refinement, and a compact ensemble can improve continuous fall localisation over published challenge baselines. Candidate motion bursts are detected first, and then their boundaries are refined using local energy traces. It then ranks candidate intervals according to features that describe motion, surrounding activity, changes in static profile, and distribution of received energy across radars and physical range bins. This choice of signal-derived features and processing operations is called physics-guided . The method does not enforce a physical differential-equation constraint on model training.

The contributions are threefold . We first describe an explicit proposal-and-scoring pipeline, where temporal parameters and feature definitions can be inspected separately from its classifiers. Second, in the computation of spatial features, we consider block-wise in-phase and quadrature representation. Third, we validate the final predictions with the official event-matching procedure and report the detection and localisation metrics. The validation predictions achieve a mean F1 of 0.8172, compared with 0.3151 for the official energy baseline. These are validation results after parameter selection, not an estimate of the performance on the la-

belled test. The analysis also shows limitations in synthetic concatenation, recording context, annotation uncertainty, and the use of future information.

2 Related literature

While prior research has established the foundation for fall detection using UWB radar, the detection and localisation of falls in continuous streams present additional complexities. In this section, we review relevant approaches on radar-based fall recognition, temporal localisation and ensemble learning, followed by evaluation strategies that inform the scope of the present study.

2.1 UWB radar fall recognition

Maitre et al. [6] investigated a CNN–LSTM architecture on measurements of three UWB radars in an apartment. Their work provides a relevant connection between temporal signal modelling and radar-based fall recognition. Imbeault-Nepton et al. [5] compared a classical feature-based approach with a convolutional approach, focusing on preprocessing and subject-wise evaluation. In sum, these studies justify the investigation of both learned representations and engineered signal summaries rather than assuming that either family is universally preferable .

Abudalfa and Bouchard investigated transfer learning with computer-vision models [3] and a hybrid vision-transformer and residual-network model [2].The approaches presented show how UWB measurements can be encoded in neural architectures as alternatives. But the binary-recognition protocols associated with this are different from the present continuous-stream task. Therefore, their reported accuracy or classification F1 scores are not used as directly comparable baselines for event localisation.

2.2 Temporal localization and ensemble scoring

Temporal action localization explicitly estimates action intervals. For example, ActionFormer combines multiscale representations and local self-attention to predict action categories and boundaries in video [8]. This creates a nice task-level distinction between recognition and localisation, however its visual representations and evaluation datasets are not UWB streams. We do not infer that its performance transfers to radar, nor claim superiority without a controlled implementation on the same benchmark.

The present approach separates interval construction from classification.Its ensemble includes extremely randomized trees, which introduce randomization in tree construction [4], alongside histogram gradient boosting and a small neural classifier. The key contribution is the incorporation of radar-informed interval features, explicit boundary refinement, and event-level evaluation on the challenge streams . Causal benefits at the component level remain to be established through controlled ablations .

2.3 Evaluation design in longitudinal sensing

Evaluation protocols should reflect the intended use of a sensing system. Opoku Asare et al. [7] investigated longitudinal smartphone and wearable measurements for depression-status prediction. Such longitudinal sensing studies provide context for participant- and day-aware evaluation, but their tasks and metrics differ from fall localization. The present experiments use the official fixed training and validation streams with event-level tIoU matching; they do not implement leave-one-participant-day-out (LOPDO) cross-validation. Citing longitudinal evaluation literature therefore does not change the experimental protocol reported here.

3 Method

The proposed method formulates continuous fall localization as a sequence of candidate generation, boundary refinement, and event scoring.The radar signals are processed to extract motion and contextual features that are evaluated by a hybrid ensemble to detect fall intervals. The four main stages are shown in Figure 1. The implementation additionally applies a final duration-dependent boundary correction after scoring and NMS; this operation is described explicitly below.

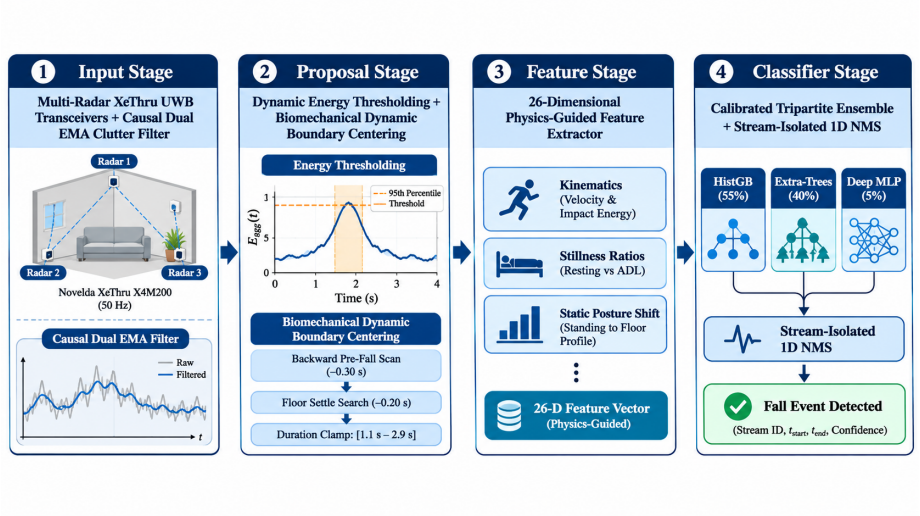


Figure 1: Overview of the proposed fall detection pipeline.

3.1 Research design and dataset

This is a computational study using the ABC2027 challenge materials [1] and UWB recordings described in the dataset literature [6]. The recordings originate from the approximately 40 m^2 LIARA apartment and use three Novelda XeThru X4M200 radars. Frames are aligned to a common 50 Hz time grid. Each HDF5 stream contains a tensor of shape $(3, T, 368)$.

The 368 values are 184 in-phase components followed by the corresponding 184 quadrature components, rather than 368 separate physical range bins.

The public training split contains 13 streams spanning approximately 6.18 hours, and validation contains five streams spanning approximately 2.05 hours. There are 359 training events and 119 validation events in the challenge annotations. Under the flagged-event evaluation policy, 186 training events and 67 validation events are evaluable, while 173 and 52 events, respectively, are ignored. These counts distinguish total annotations from events contributing to the evaluation denominator. The challenge documentation describes five unlabeled test streams spanning approximately 2.07 hours. No labeled test score is available in this study.

Published streams are artificial concatenations with 0.2 s crossfade at joins. A fall is defined as the initiation of movement and ends when the person is stable on the floor; subsequent lying is not included. The challenge documentation quotes a boundary uncertainty of $\approx \pm 0.5$ – 1 s. Within each source corpus, subjects are split, but there is no link between fall and daily-activity corpus subject identities.

3.2 Signal representation and background removal

Let $h_{r,n,c}$ denote real-valued component c from radar r at frame n. For physical range bin $k \in \{0, \ldots, 183\}$, the complex signal is

$$z_{r,n,k} = h_{r,n,k} + \mathrm{i} h_{r,n,k+184}. \tag{1}$$

Background subtraction is applied independently to each stored component. With frame interval $\Delta t = 0.02$ s, the exponential background and residual are

$$\begin{aligned} \alpha_\tau &= \exp(-\Delta t/\tau), \\ b_{r,n,c} &= \alpha_\tau b_{r,n-1,c} + (1-\alpha_\tau) h_{r,n,c}, \\ u_{r,n,c} &= h_{r,n,c} - b_{r,n,c}. \end{aligned} \tag{2}$$

The initial background equals the first frame. The implementation computes residuals at $\tau = 0.8$ and 3.0 s. However, proposal generation and the returned candidate features use the fast residual; the medium residual is computed but does not directly enter the final 26-dimensional representation. Therefore, the results reported do not demonstrate any benefit of dual-scale filtering.

The aggregate trace and per radar power for the fast residual are

$$\begin{aligned} P_{r,n} &= \sum_{c=0}^{367} u_{r,n,c}^2, \\ \widetilde{P}_{r,n} &= \frac{P_{r,n}}{\mathrm{median}_m\, P_{r,m} + 10^{-7}}, \\ E_n &= \max_r \widetilde{P}_{r,n}. \end{aligned} \tag{3}$$

The sum of the squared I/Q values provides a measure of total power per bin. The max function ensures that, if there are variations in the

responses of the radars, the response of just one radar will be selected. This helps improve robustness to non line of sight scenarios.

A temporal-change trace is also computed:

$$\begin{aligned} D_n &= \frac{1}{3}\sum_{r=1}^{3}\sum_{c=0}^{367}(h_{r,n,c} - h_{r,n-1,c})^2, \\ \widetilde{D}_n &= \frac{D_n}{\operatorname{median}_m D_m + 10^{-7}}. \end{aligned} \tag{4}$$

with $D_0 = 0$. This does not represent a true physical measure of velocity/acceleration but simply quantifies CIR variability. Although the background filter is causal, medians are taken across the whole stream making this feature and subsequent features non-causal and offline.

3.3 Proposal generation and boundary refinement

Active frames satisfy $E_n \geq Q_{95}(E)$, where Q_{95} is the stream's 95th percentile. Consecutive active frames form fragments. Fragments are merged when their separation is at most 0.6 s, or at most 1.2 s if the mean intervening aggregate energy exceeds 3.0. Merged fragments shorter than 0.20 s or longer than 6.0 s are discarded.

For each retained fragment, let n_p be its maximum-energy frame and $E_p = E_{n_p}$. The onset and settling thresholds are

$$q_s = \min(15, \max(2, 0.03E_p)), \qquad q_e = \min(30, \max(3.5, 0.04E_p)). \tag{5}$$

The algorithm searches backward by up to 2.5 s and selects the last frame at or below q_s. The proposed start is that time minus 0.30 s, clipped at zero. If no such frame exists, the start defaults to 1.30 s before the peak. A forward search of up to 2.2 s selects the first frame at or below q_e. Its time minus 0.20 s gives the initial end, subject to a minimum duration of 1.1 s. If no settling crossing exists, the end defaults to 1.0 s after the peak.

For an interval of less than 1.1 s, the start is shifted back by half of the duration deficit and the end is extended to a 1.1 s interval. Intervals longer than 2.9 s are truncated to 2.9 s from the proposed start time. Final bounds are clipped to the stream duration, so boundary-adjacent intervals can be shorter than the nominal minimum. These are empirical parameters, not universal physiological constraints.

3.4 Candidate features

Each candidate $[s, e]$ is represented by 26 features, summarized in Table 1. Event features use samples between its start and end. Context windows are $[s-1.5, s-0.2]$, $[e+0.2, e+1.5]$, and $[e+1.5, e+3.0]$ seconds, clipped to available samples using the implementation's index rules. Context-to-event energy ratios use an additive denominator constant of 10^{-6}.

For the spatial features, component powers must be paired before as-

Table 1: Candidate features in the implemented representation. Indices are zero-based.

Indices	Features
0–4	Duration; aggregate-energy peak and mean; normalized CIR-difference mean and peak.
5–10	Pre-event, immediate post-event, and extended post-event mean energies; each divided by event mean energy.
11–13	Three per-radar shares of mean fast-residual event power.
14–17	Pre/post raw-CIR profile difference: joint Euclidean norm, maximum absolute component, maximum per-radar norm, and cosine distance.
18–19	Energy-weighted centroid and spread across 184 paired complex range bins.
20–25	Aggregate-energy skewness, excess kurtosis, fraction of the peak position, peak contrast, radar-share entropy, median normalised radar power over radar/frame entries.

signing a bin position:

$$
\begin{aligned}
w_k &= \sum_r \sum_{n\in[s,e]} (u_{r,n,k}^2 + u_{r,n,k+184}^2),\\
c &= \frac{\sum_k k w_k}{\sum_k w_k + 10^{-8}},\\
\sigma &= \sqrt{\frac{\sum_k (k-c)^2 w_k}{\sum_k w_k + 10^{-8}}}.
\end{aligned}
\tag{6}
$$

The centroid and spread are measured in units of bin indices, not in units of meters. Static-profile features compare the mean raw CIR in the pre-event and immediate post-event windows . Energy-share entropy uses normalised shares with small numerical differences. The final feature is the median over all normalised radar/frame entries in the candidate, not the median of the max-aggregated trace. These features characterise temporal activity, changes in the static CIR profile, and spatial distribution of residual power together.

3.5 Training and ensemble inference

A training proposal is positive when its maximum tIoU with an evaluable fall is at least 0.25. Otherwise, proposals overlapping an ignored event expanded by $\pm 2\,\mathrm{s}$ are excluded from fitting; remaining proposals are negative. Evaluable matches take precedence over this exclusion. All validation proposals are retained for inference and evaluation. Validation labels are used only for operating-point selection and, if requested, subsequent final refitting.

The candidate score is

$$
S(\mathbf{x}) = 0.55 S_{\mathrm{HGB}}(\mathbf{x}) + 0.40 S_{\mathrm{ET}}(\mathbf{x}) + 0.05 S_{\mathrm{MLP}}(\mathbf{x}). \tag{7}
$$

Histogram gradient boosting uses 180 maximum iterations, learning rate 0.03, at most 20 leaves per tree, and a minimum leaf size of eight. Extra-Trees uses 180 trees, maximum depth 12, and minimum leaf size four [4]. Both tree classifiers use balanced class weights. They receive the unstandardized feature vectors.

The MLP receives features standardized using statistics fitted on training proposals. It has two 64-unit hidden layers, each followed by batch normalization, GELU, and dropout with probability 0.20. A single output logit is converted to a score using a sigmoid. Training uses positively weighted binary cross-entropy and AdamW with weight decay 10^{-2}. The implementation defaults are 15 epochs, learning rate 10^{-3}, and batch size 32. The neural output contributes to the final ensemble score alongside the two tree classifiers.

The implementation searches score thresholds from 0.18 through 0.44 in increments of 0.01, using mean validation F1 and the challenge tie-breakers. Thresholded proposals undergo greedy NMS at tIoU 0.20, independently for each stream. During test inference, neither scaler fitting nor cross-stream statistics are performed. No matching of the contents of the repeated recording is implemented. Whole-stream normalisation and future context are calculated only within the current stream.

3.6 Final boundary centering and submission export

The implementation proposed separates the intervals for feature extraction and scoring from the intervals finally written to the submission CSV. After thresholding and stream-wise NMS, the export function applies an empirical correction to each remaining interval $[s, e]$.With its original duration $d = e - s$, the corrected bounds are

$$\begin{aligned} s_f &= s + 0.19\,\mathrm{s}, \\ \delta_e &= 0.08\,\mathrm{s} + 0.30\big(d - 1.8\,\mathrm{s}\big), \\ e_f &= \max\big(s_f + 0.90\,\mathrm{s},\ e - \delta_e\big). \end{aligned} \tag{8}$$

The beginning is put back 0.19 s, and the end adjustment is dependent on the duration of the original proposal. For d less than about 1.533 s, δ_e is negative, so this operation can lengthen, rather than shorten, the end. So it is a time-dependent temporal adjustment and not a globally compressive damping operation. This final step has a 0.90 s minimum, which is different from the nominal 1.1 s minimum at the time of the proposal construction.

Features and confidence scores are not recomputed on $[s_f, e_f]$, and NMS is not repeated after this adjustment. The writer checks the final values against stream durations before official CSV validation. The parameters are empirical temporal adjustments rather than direct measurements of biomechanical transitions or electromagnetic multipath decay. The correction is applied once to each retained interval. All reported final results below refer to the serialized CSV after this operation.

3.7 Evaluation protocol

Predictions contain a stream identifier, start time, end time, and score. For intervals A and B, temporal intersection-over-union is

$$\mathrm{tIoU}(A, B) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}. \tag{9}$$

The official evaluator processes predictions in order of descending score, and greedily matches each to the highest-overlap available evaluable event that satisfies threshold θ. Each event is matched at most once. Only after evaluable matching are unmatched predictions overlapping expanded ignore zones removed. Remaining unmatched predictions and events count as false positives and false negatives.

The ranking metric and false-alarm rate are

$$\overline{F}_1 = \frac{F_1(0.1) + F_1(0.3) + F_1(0.5)}{3}, \qquad \mathrm{FA/h} = \frac{\mathrm{FP}}{\sum_j T_j / 3600}. \tag{10}$$

F1 at 0.7 is additionally reported. Ranking ties use median absolute start error at 0.3 and then false alarms per hour at 0.3. Flagged events are handled with the test-like ignore policy. Boundary tolerances of 0.5 and 1.0 s are descriptive measures and do not relax tIoU matching.

4 Results and Discussion

The proposed pipeline was evaluated on its ability to detect falls and estimate their temporal boundaries within continuous radar streams. Event F1 at multiple temporal overlap thresholds, boundary estimation errors, and false alarms per hour provide complementary measures of detection reliability and localization accuracy.

4.1 Validation performance

The official ABC2027 event-level scoring procedure was used for evaluation, where flagged events were treated as ignore zones. The five validation streams cover around 2.05 h and contain 67 evaluable autumn events. The final validation output has 63 predicted intervals and achieves a mean event F1 of 0.8172 at tIoU thresholds of 0.1, 0.3 and 0.5.

The comparison between the proposed method and the official challenge baselines is reported in Table 2. It gives a mean F1 that is 0.5021 better than the energy baseline, which is an improvement of about 159.4% relative. The energy-baseline results are consistent with the official evaluation report, and the logistic-regression results are the rounded values reported by the challenge. These are benchmark-specific validation comparisons and are not evidence of superiority across datasets or experimental protocols.

The method identified 52 true positives, produced five false positives, and missed 15 evaluable falls at the primary threshold of tIoU is 0.3. Six predictions overlap ignore zones without matching an evaluable event and are not scored. Precision is 0.9123, recall is 0.7761, false-alarm rate is 2.44

Table 2: Validation event F1 with flagged events ignored. Logistic-regression values are rounded challenge documentation values.

Method	Mean F1	F1@0.1	F1@0.3	F1@0.5	F1@0.7
Logistic baseline	0.258	0.593	0.167	0.014	0.000
Energy baseline	0.3151	0.3582	0.3383	0.2488	0.0597
Proposed method	0.8172	0.8387	0.8387	0.7742	0.4839

per hour. These results show better suppression of non-fall detections than the energy baseline with a false alarm rate of 48.81 per hour. However, the other missed events and false alarms still are relevant limitations for continuous surveillance.

Table 3: Event-level detection results across temporal overlap thresholds.

θ	TP	FP	FN	Precision	Recall	F1	FA/h
0.1	52	5	15	0.9123	0.7761	0.8387	2.44
0.3	52	5	15	0.9123	0.7761	0.8387	2.44
0.5	48	9	19	0.8421	0.7164	0.7742	4.39
0.7	30	27	37	0.5263	0.4478	0.4839	13.18

4.2 Temporal localization accuracy

Mean tIoU = 0.7001 for matched events at $\theta = 0.3$. The mean absolute start and end errors are 0.3642 and 0.3694 s respectively, and the median errors are 0.18 and 0.29 s. Start estimates are within ± 0.5 s for 71.15% of matched events and within ± 1.0 s for 92.31%. End estimates meet these tolerances at 78.85% and 96.15%, respectively. These percentages describe separate start and end errors among matched events, not joint boundary agreement over all annotated falls.

The performance drops as the required temporal overlap becomes more strict. F1 is 0.8387 at tIoU 0.3, 0.7742 at 0.5 and 0.4839 at 0.7. At the most stringent threshold, there are 30 matched events, 27 false positives and 37 unevaluated events. Thus, the pipeline is successful in identifying many fall intervals, but precise alignment of both boundaries is still challenging. Although the reported annotation uncertainty of roughly ± 0.5–1.0 s provides some perspective to these errors, it does not change the official matching criterion.

The duration-dependent correction in Equation 8 is included in the final localisation procedure. Features and confidence scores are computed on the candidate intervals, while the corrected intervals are used for the final evaluation. The results presented here characterise this entire pipeline; they do not isolate the contribution of the correction or individual feature groups.

The output of the prediction does not contain `t_alarm`. Therefore, the evaluator predicts the latency field of start minus annotated start. Its median value of -0.06 s describes signed placement of the onset. This

value does not represent measured real-time alarm latency because the method uses whole-stream normalisation and post-event context.

4.3 Test-stream inference

The method generates 56 predicted fall intervals across the five unlabeled test streams. The numbers of predictions are 15 for `test_00`, seven for `test_01`, 13 for `test_02`, 16 for `test_03`, and five for `test_04`. Each stream is processed independently using the same feature extraction, ensemble scoring, NMS, and final boundary correction.

The submission contains the required fields `stream_id`, `t_start`, `t_end`, and `score`. Since test annotations and a labeled test evaluation are unavailable, these counts describe model predictions rather than confirmed falls. Test precision, recall, and ranking F1 therefore cannot be inferred from the submission file.

4.4 Limitations and future evaluation

The benchmark contains simulated falls at fixed mat locations and daily activities recorded in different apartment contexts.Thus, static CIR changes and context features of energy can capture conditions of recording and fall dynamics. The assessment should be conducted in independent rooms and with naturally occurring falls to determine generalisation beyond this setting.

Only 67 evaluable falls in the validation split are included in the flagged-event policy. The reported score is further a tuned validation result and not an unbiased estimate of test performance due to parameter selection on this split. The evaluation does not include repeated-run confidence intervals, controlled component ablations, or comparison to an external dataset.

The current version is offline because it uses whole-stream statistics and future context. However, conclusions about real-time or edge deployment, runtime, peak memory and actual alarm-response measurements are needed before making any conclusions. Future work should also explore proposal recall in isolation from classification performance and calibrate boundary adjustments on data separate from final evaluation.

5 Conclusion

The paper presents a physics-guided pipeline for detecting and temporally locating falls in continuous multi-radar UWB streams. The method combines energy-based proposal generation, 26-dimensional feature extraction, hybrid ensemble scoring and duration-dependent boundary adjustment. Its decoupling of the candidate construction, classification and localisation results in an interpretable continuous fall monitoring approach.

On the ABC2027 validation split, the method achieved a mean event F1 of 0.8172 across tIoU thresholds of 0.1, 0.3, and 0.5. At the primary threshold of 0.3, precision reached 0.9123 and recall reached 0.7761, with 2.44 false alarms per hour. Mean tIoU was 0.7001, while median absolute

start and end errors were 0.18 and 0.29 seconds, respectively. These results demonstrate stronger validation performance than the official challenge baselines.

Nevertheless, generalisation is limited by simulated falls, synthetic stream construction, and parameter selection based on validation. The use of whole-stream statistics and future context also means that the results cannot be interpreted as real-time alarm performance. Future work should consider independent environments, naturally occurring falls, controlled component ablations, and causal processing with measured computational and alarm latency. Such an evaluation is needed to assess the suitability of the method for practical use.

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